

Teaching Statement

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Teaching philosophy. My main goal as an instructor is to make the structure behind a mathematical idea visible. Complicated arguments should be built from simple and concrete ideas, and abstract concepts should be connected, whenever possible, to examples that students can compute or visualize. I want students not only to know how to apply a theorem, but also to understand why its hypotheses are natural, what phenomenon a definition is designed to capture, and how a difficult argument can be broken into manageable steps.

My teaching experience ranges from introductory undergraduate courses to graduate mathematics. At UC Santa Barbara, I taught problem-solving sections in calculus, vector calculus, and transition-to-higher-mathematics courses, and I served as a mentor in the Directed Reading Program. At Northeastern University, I have taught Calculus II for Science and Engineering, undergraduate Linear Algebra, and a graduate course in Morse Theory. Teaching at these different levels has reinforced my belief that students learn difficult mathematics most effectively when new ideas grow naturally from familiar ones.

Examples, visualization, and computation. In my undergraduate courses, I use concrete examples to motivate abstract definitions before developing the general theory. In Linear Algebra, for example, I do not want students to encounter linear independence, diagonalization, or inner products only as formal definitions. I first work through examples that reveal the underlying phenomenon and then use the formal theory to explain what the examples have in common. Students have particularly appreciated the applications, worked examples, and review materials in my courses; one student wrote that I ‘gave cool applications’ and another emphasized that I ‘took [my] time to explain topics which made understanding them easier.’

I also use technology when it can make an abstract idea visible. I write short Python programs for my courses that allow students to experiment directly with mathematical concepts. In Linear Algebra, I use singular value decomposition to compress an image and let students see how retaining different numbers of singular values changes the quality of the reconstructed image. This gives a concrete interpretation of low-rank approximation and makes the role of singular values much more tangible than the matrix formulas alone.

I have also used programs in calculus and Fourier analysis. For example, an interactive program illustrates how the secant line through x and $x + \Delta x$ approaches the tangent line as $\Delta x \rightarrow 0$, making the definition of the derivative visible before introducing the formal limit. Another program displays how Fourier partial sums approximate a given function as more terms are added. I do not use computation as a substitute for mathematical reasoning; rather, I use it to help students recognize the mathematical structure that the subsequent definitions and proofs describe.

At the graduate level, I place greater emphasis on the architecture of proofs. In my Morse Theory course, I tried to make clear not only the individual technical arguments but also why the main constructions are introduced and how the different parts of the theory fit together. A student described me as ‘an expert in the course material’ who ‘could articulate all the details fairly clearly.’ My goal is for graduate students to leave a course with a conceptual map of the subject, rather than only a collection of individual theorems.

Organization and continuous improvement. I regard organization and transparency as essential parts of teaching. Homework, quizzes, review materials, and examinations should reinforce one another, and students should have a clear understanding of what they are expected to learn. I regularly work through representative problems, provide structured review materials, and make the intermediate steps of difficult calculations explicit. A student in Linear Algebra wrote that the course materials prepared students well for examinations, so that “nothing felt too surprising and unfair.”

An important part of my development as an instructor has been learning to use student feedback systematically. During my first year teaching at Northeastern, students appreciated the amount of practice I provided but also asked for simpler language, more examples, clearer problem-solving steps, and a slower

pace on difficult topics. I responded by organizing lectures more explicitly around their central learning goals, increasing the number of worked examples, writing intermediate steps more carefully on the board, and developing more structured review materials.

These changes were reflected in subsequent courses. By Fall 2025, my Linear Algebra sections received overall teaching-effectiveness ratings above the corresponding departmental average, and the large majority of instructor-related categories were also above departmental averages. The written comments from these courses repeatedly emphasized examples, applications, organization, review materials, and accessibility. I regard this improvement not as a finished process, but as evidence that carefully listening to students and adjusting the way I communicate mathematics can make a substantial difference.

Supporting students and mentoring. I want students to feel comfortable asking questions, including questions about points that may initially seem elementary. Office hours are an important extension of the classroom, particularly for students who need more time to develop confidence with abstract mathematics. I also try to combine clear academic expectations with reasonable flexibility when students face temporary difficulties, and I work carefully with students receiving academic accommodations while respecting their privacy. The goal is not to lower mathematical standards, but to make sure that circumstances unrelated to mathematical understanding do not unnecessarily prevent a student from demonstrating what they have learned.

Beyond the classroom, I value opportunities to help students move from coursework toward independent mathematics. My experience with the Directed Reading Program, graduate teaching, and individual student mentoring has made me particularly interested in reading courses and undergraduate research projects. Depending on a student's background, topics in spectral theory, geometry, or mathematical physics can often be approached through concrete starting points—for example, eigenvalues of elementary differential operators, Fourier analysis, or the geometry of surfaces—before moving toward more advanced ideas.

Ultimately, I want students to develop the habit of asking why a definition is natural, how a theorem can be anticipated from examples, and how a complicated problem can be decomposed into simpler ones. At the undergraduate level, I hope this gives students confidence in confronting abstract mathematics; at the graduate level, I hope it helps them begin to recognize the ideas that organize a subject beyond individual proofs. I will continue to refine my teaching with the goal of making mathematics rigorous and challenging without making it unnecessarily inaccessible.

Selected student feedback

- [1] ‘Gave cool applications.’ ‘Worked through example problems.’ ‘Super helpful review sheets.’ — Students from MATH 2331 Linear Algebra, Northeastern University, Fall 2025.
- [2] ‘The instructor was very good at giving us examples to work through and took his time to explain topics which made understanding them easier.’ — MATH 2331 Linear Algebra, Northeastern University, Fall 2025.
- [3] ‘The instructor is an expert in the course material and could articulate all the details fairly clearly.’ — MATH 7371 Morse Theory, Northeastern University, Fall 2025.
- [4] ‘The course material was very well suited to prepare you for the exams. Nothing felt too surprising and unfair come the exam.’ — MATH 2331 Linear Algebra, Northeastern University, Fall 2025.
- [5] ‘I went to the office hours a few times, the instructor is very willing to help, and it was very effective.’ — MATH 2331 Linear Algebra, Northeastern University, Fall 2025.